https://ijnm.co.in/

Indian Journal of Nuclear Medicine



*Pictorial Essay*

# Demystifying Cardiac Infections with Nuclear Medicine

Shrikant Vasantrao Solav[1], Rajlaxmi Rangrao Jagtap[1], Shailendra Vasant Savale[1]

[1]Department of Nuclear Medicine, Spectlab Nuclear Medicine Services, Bavdhan, Pune, Maharashtra, India.

## ABSTRACT

Infections in relation to the heart could be localised, as in a mycotic aneurysm, or systemic (as in infective endocarditis with septicemia). Echocardiography remains a mainstay in diagnosing native valve endocarditis. However, it has limitations in prosthetic valve endocarditis. An infected cardiac device is another challenge for direct visualisation of the focus of infection, and computed tomography (CT) and magnetic resonance imaging have limited application other than the detection of structural changes such as an aneurysm. Fluorine 18-fluorodeoxyglucose positron emission tomography-computed tomography is a sensitive tool to detect a focus of infection/inflammation, but its use in detecting myocardial inflammation involves stringent preparations such as prolonged fasting and preadministration of heparin. Gallium-68-fibrinogen activation protein inhibitor is a recently introduced molecule which overcomes this difficulty, as it does not show any uptake in the normal heart. It does not require fasting status or stringent control of blood sugar. It does localise to the site of infarction, and there is limited literature related to its use in infections involving the heart. This report presents six cases with clinical suspicion of infections involving stents, the endocardium, and cardiac implantable electronic devices, where nuclear medicine procedures played a crucial role in influencing the diagnosis.

**Keywords:** Cardiac infections, Fibrinogen activation protein inhibitor positron emission tomography–computed tomography, Fluorodeoxyglucose positron emission tomography–computed tomography, Mycotic aneurysm

## INTRODUCTION

Infections involving the heart can affect the endocardium, native and prosthetic valves, stents, and cardiac implantable electronic devices (CIEDs). Blood culture and transesophageal echocardiography (TEE) are fundamental diagnostic tools; however, prior antibiotic exposure and the presence of prosthetic valves and implantable devices often complicate diagnosis using these methods.

Fluorine 18-fluorodeoxyglucose (FDG) positron emission tomography (PET)–computed tomography (CT) is a well-established imaging modality for detecting infections across various body regions, including the heart. However, its unpredictable physiological uptake and stringent preconditions – such as blood sugar regulation and prolonged fasting – pose challenges in accurate diagnosis. Gallium-68-fibroblast activation protein inhibitor (FAPI) PET-CT offers a promising alternative, overcoming these limitations as it does not require fasting or strict blood sugar control.

This pictorial essay presents a range of clinical scenarios involving cardiac infections, highlighting the role of advanced nuclear medicine techniques in diagnosis.

## MATERIAL AND METHODS

Six patients with clinically suspected cardiac infections – including stent-related infections, infective endocarditis, and infections involving CIEDs – were included in the study.

- Myocardial perfusion studies were performed in five

***Corresponding author:** Dr. Shrikant Vasantrao Solav, Sr. No. 268, Mumbai, Pune Banglore Highway Pashan Exit, behind Crystal Honda Showroom, Bhunde Vasti, Bavdhan, Pune - 411 021, Maharashtra, India. drsolav@gmail.com

**Received:** 15 May 2025 **Accepted:** 22 December 2025 **Published:** 10 March 2026 **DOI:** 10.25259/IJNM_62_25

This is an open-access article distributed under the terms of the Creative Commons Attribution-Non Commercial-Share Alike 4.0 License, which allows others to remix, transform, and build upon the work non-commercially, as long as the author is credited and the new creations are licensed under the identical terms. ©2026 Published by Scientific Scholar on behalf of Indian Journal of Nuclear Medicine

patients [Fig 1-5]

- FDG PET-CT scans were conducted in two patients [Fig 1 and 3]
- Ga-68 FAPI PET-CT scans were performed in three patients [Fig 2, 4, and 6]
- One patient underwent both FDG and FAPI PET-CT scans [Fig 5].

Myocardial perfusion studies were conducted at rest using an intravenous administration of 5-7 mCi of technetium-99m sestamibi. Gated single-photon emission computed tomography (SPECT) images of the myocardium were acquired 30–40 min postadministration using a Siemens Evo Excel dual-head gamma camera system with a low-energy, high-resolution collimator. The images were reconstructed and displayed alongside FDG and FAPI PET-CT images.

FDG PET-CT scans were performed using 5–9 mCi of FDG administered intravenously in three patients following a prolonged fasting period of 14 h. Imaging was conducted 1 h postinjection using a Siemens Horizon PET-CT system.

Gallium-68 FAPI PET-CT imaging, an intravenous dose of 3–4 mCi of gallium-68 FAPI, was administered to four patients, and imaging was performed 1 h postinjection.

## RESULTS

All patients presented with fever and clinical signs of bacteremia/septicemia. Leukocytosis was observed in four patients, and blood cultures identified microbial pathogens in three cases [Table 1].

Case 1: Perfusion-metabolism mismatch was observed in the basal anteroseptal segment [Fig 1b and c], with FDG uptake corresponding to the stent [Fig 1d, e, h, and i].

Case 2: FAPI uptake was noted along the right coronary artery (RCA) stent [thin arrow in Fig 2g-j]. Additionally, a perfusion defect in the anteroseptal and apical myocardium [Fig 2e] with corresponding FAPI uptake was associated with a recent myocardial infarction [thick arrow in Fig 2j and k].

Case 3: A perfusion defect in the basal inferolateral segment was identified [thick arrow in Fig 3b] accompanied by FDG uptake along the circumflex stent [curved arrow in Fig 3f, h, and j] and an associated pericardial reaction, as seen in PET-CT images.

Case 4: FAPI localisation was found along the RCA stent [thin arrows in Fig 4c-f, i, and j], distinct from recent inferior wall infarct-related uptake [thick arrows in Fig 4h and l]. This patient did not undergo myocardial perfusion imaging (MPI).

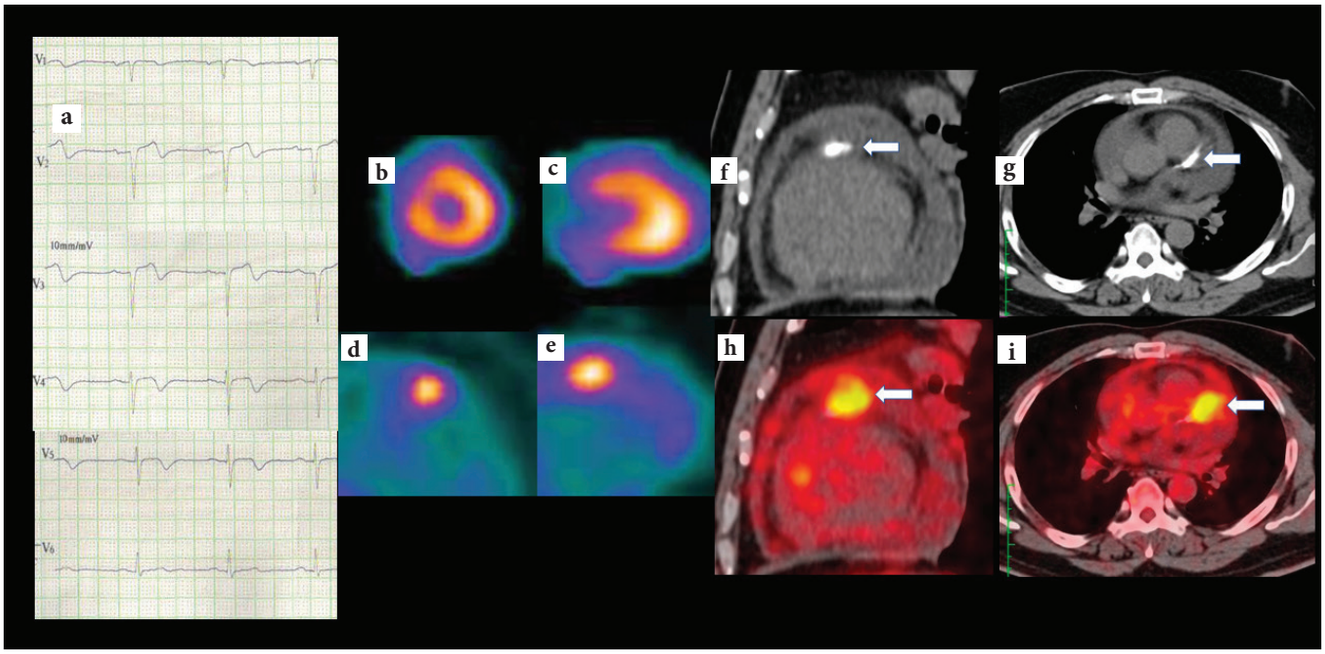



**Fig 1:** (a-i) A 49-year-old man had angioplasty to the left anterior descending (LAD) following an acute anterior wall infarct. At 2 weeks postpercutaneous transluminal coronary angioplasty, he developed chest pain. Hemoglobin – 16.1 g/dl, white blood cell – 19,400/ml, and erythrocyte sedimentation rate 42 mm/1$^{st}$ h. Repeat angiogram showed aneurysmal dilatation of the LAD near the stent with thrombus in the proximal part. (a) Electrocardiogram showed anterior wall infarct, (b and c) Myocardial perfusion imaging at rest shows mild perfusion defect in basal anteroseptal myocardium, (d and e), Fluorodeoxyglucose (FDG) positron emission tomography (PET) scan showed focal hypermetabolism in the basal septal myocardium. (f,g,h and i) Computed tomography (CT) and PET-CT fusion images (White arrows) showed FDG uptake corresponding to the hyperdense stent in LAD. corresponding to the aneurysmal dilatation on angiogram (not shown here)

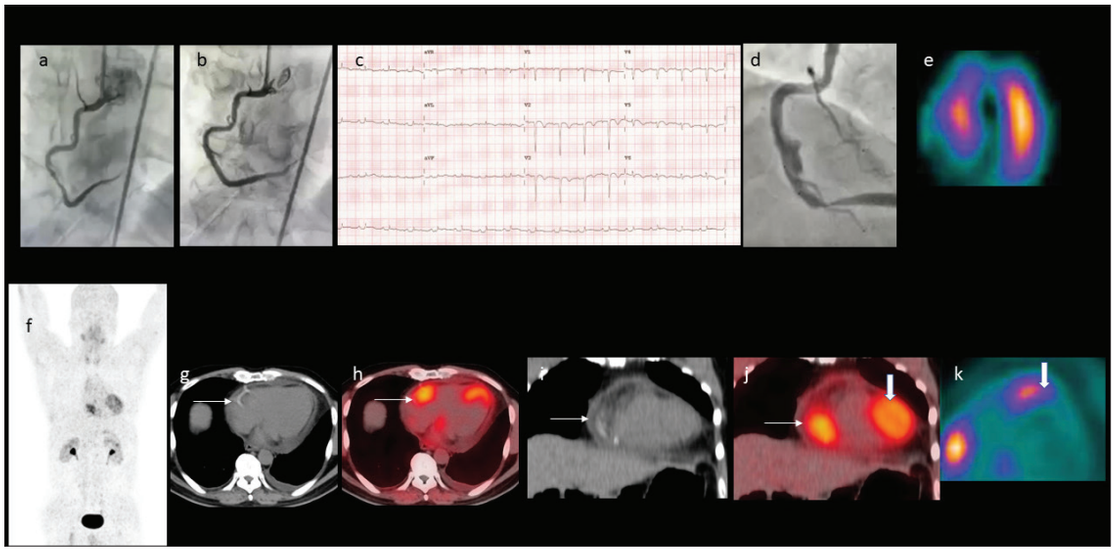



**Fig. 2:** (a-k) A 55-year-old man had an inferior wall infarct. Angiogram revealed lesions in left anterior descending (LAD) and right coronary artery (RCA). Percutaneous transluminal coronary angioplasty (PTCA) was done for LAD-RCA and POBA to PLV branch 9 days postinferior wall infarct. (a and b) Pre- and post-PTCA RCA lesion is depicted. Postprocedure fever and chest pain lasted for 14 days. Echocardiogram showed left ventricular ejection fraction of 45% with hypokinetic anteroseptal and apical myocardium suggesting anteroseptal infarct in the intervening phase. (c) Electrocardiogram at the second admission revealed an inferior and anteroseptal myocardial infarction. White blood cell was 11410/ml, NT-proBNP 396.3 pg/ml (<125), and troponin T 27.27 pg/ml (<17). (d) A repeat angiogram showed restenosis of RCA with proximal aneurysmal dilatation of RCA. (e) Myocardial perfusion imaging showed apical-distal septal perfusion defect. (f) Fibrinogen activation protein inhibitor positron emission tomography computed tomography scan showed increased tracer localization in two regions of the heart on maximum intensity projection. (Horizontal thin arrow in g, j) Tomogram showed increased tracer localization along the stent in RCA suggesting mycotic aneurysm. (Vertical thick arrow in j and k) Tracer uptake in the anterior wall corresponded with recent infarction.

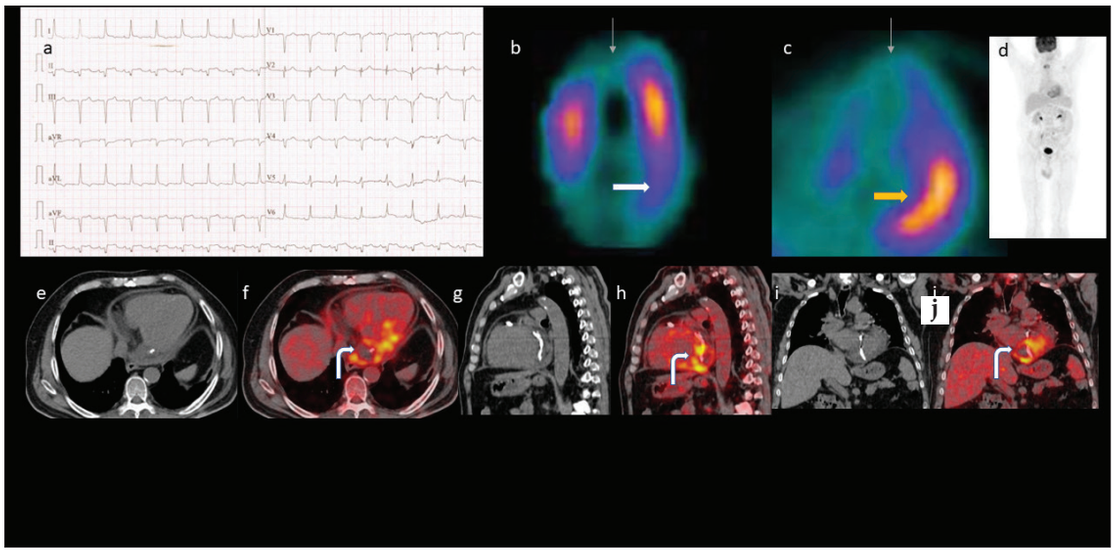



**Fig. 3:** (a-j) A 63-year-old man presented with cough and dyspnea. (a) Electrocardiogram revealed an old anterior wall infarct and recent inferior wall infarct. There was a history of percutaneous transluminal coronary angioplasty to the circumflex coronary artery 2 weeks earlier. He had a fever with chills for 10 days. Echocardiography showed an ejection fraction of 35%. Pericardial effusion was noted. Blood culture showed growth of Pseudomonas aeruginosa. Check angiogram showed aneurysmal dilatation of the circumflex coronary artery abutting the stent. (Thin arrow in b) Tc-sestamibi scan showed perfusion defects in distal septal and apical myocardium (thin arrow in b) with matched defects on (Thin arrow in c corresponding to old infarct) fluorodeoxyglucose (FDG) sca. (White solid arrow in b) The basal inferolateral myocardium showed a perfusion defect with FDG uptake (yellow solid arrow in c). (Curved arrows in f,h and j) Positron emission tomography (PET)–computed tomography (CT) showed FDG-avid collection surrounding the stent in fusion PET-CT images, (e, g, and i) corresponding plain CT images. Conclusion mycotic aneurysm

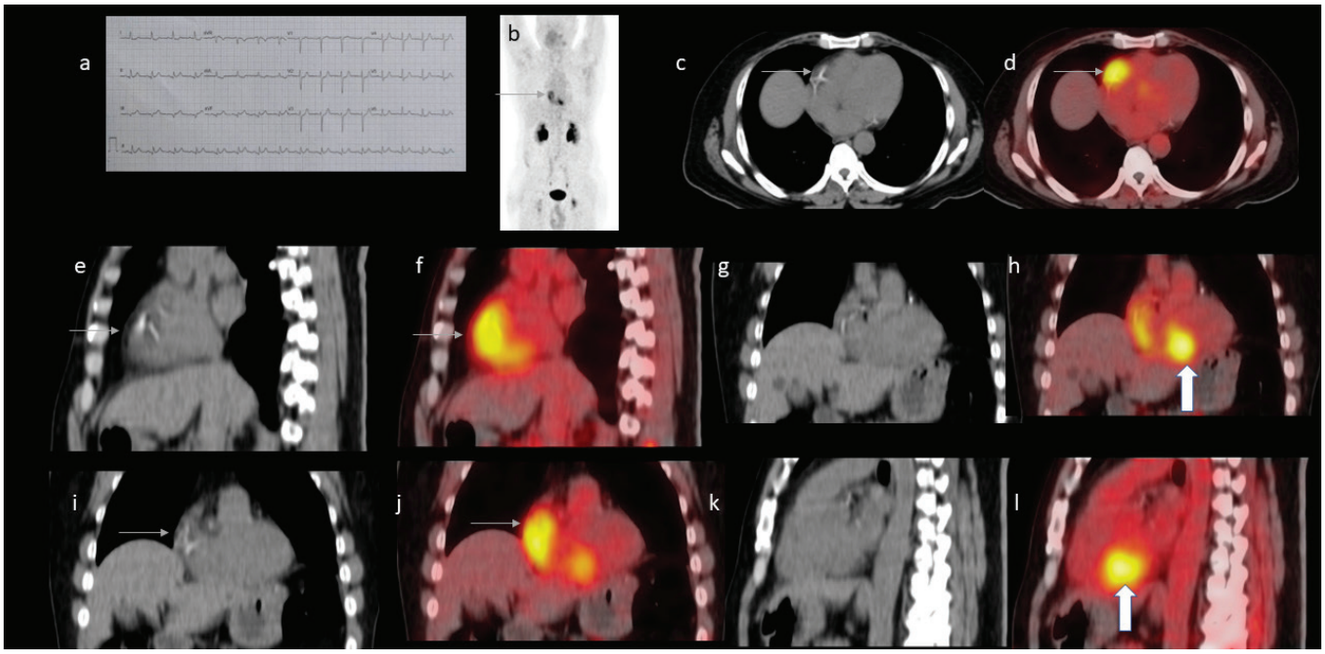



**Fig 4:** (a-l) A 68 years old gentleman with history of angioplasty to circumflex coronary artery seven years ago had recent acute coronary syndrome. Angiogram revealed significant lesion in right coronary artery for which he underwent angioplasty and stent placement. He developed high grade fever within 5 days and blood culture was positive for pseudomonas aeruginosa. His WBC was 12180/ml. CRP was 115mg/L (normal<6). ECG revealed inferior wall infarct. (a) Echocardiography showed inferior wall hypokinesia. Repeat angiography showed heavy thrombotic occlusion of RCA for which multiple thrombus suctions were performed. Ga68-FAPI PET scan showed avid tracer localisation surrounding the RCA stent compatible with infection in the vicinity- thin arrows in b (MIP); c, d (axial) (thin arrows); e, f (sagittal) (thin arrows); i, j (coronal) (thin arrows). Inferior wall tracer localisation is related to recent infarction. (thick vertical arrows in h and l.) Corresponding CT sections are also shown g and k). Patient responded to antibiotics.

Case 5: The patient had septicemia with evidence of leptomeningitis on magnetic resonance imaging [Fig 5a]. MPI revealed a mild perfusion defect in the basal inferolateral segment [thin arrow in Fig 5f]. FAPI uptake was seen distal to the stent [thick arrows in Fig 5g, h, j, k, and l] involving the inferior myocardium, with a discordant pattern between MPI and FAPI uptake. Additionally, abnormal uptake was detected in the meninges [curved arrow in Fig 5b], spleen, and both kidneys [arrowheads in Fig 5c-e].

Case 6: This patient had a cardiac synchronisation device and presented with fever and bacteremia. MPI revealed a perfusion defect, possibly related to the pacemaker lead [thin arrow in Fig 6b and c]. FDG PET was inconclusive, but FAPI PET conducted the following day showed tracer localisation along the coronary sinus lead [horizontal arrows in Fig 6e, f, h, and i]. The right ventricular lead and pulse generator exhibited no abnormal tracer uptake [Fig 6j and k], ruling out infection in these sites.

## DISCUSSION

The term mycotic aneurysm was first introduced by William Osler to describe infected aneurysms, encompassing all types of vascular infections rather than being limited to fungal causes. The incidence of coronary artery aneurysm ranges from 0.5% to 5%.[1]

Coronary artery aneurysms can arise as complications following stent placement (both bare-metal and drug-eluting stents). These aneurysms are attributed to factors such as vessel wall injury, dissection, high-pressure balloon inflation, and stent malapposition.[2]

The risk of infected coronary artery aneurysms stems from procedural vessel wall injury, transient bacteremia, direct arterial invasion, or septic embolisation.[3] The most commonly implicated organisms include *Staphylococcus aureus* and *Streptococcus haemolyticus*. Among the six cases reported here, two involved *Pseudomonas aeruginosa*, and one was caused by *Streptococcus haemolyticus*.

Culture-negative mycotic aneurysms or infective endocarditis often result from prior antibiotic treatment or infections caused by fastidious organisms. Mycotic aneurysms are pseudoaneurysms that form due to disruption of the external elastic lamina.[4]

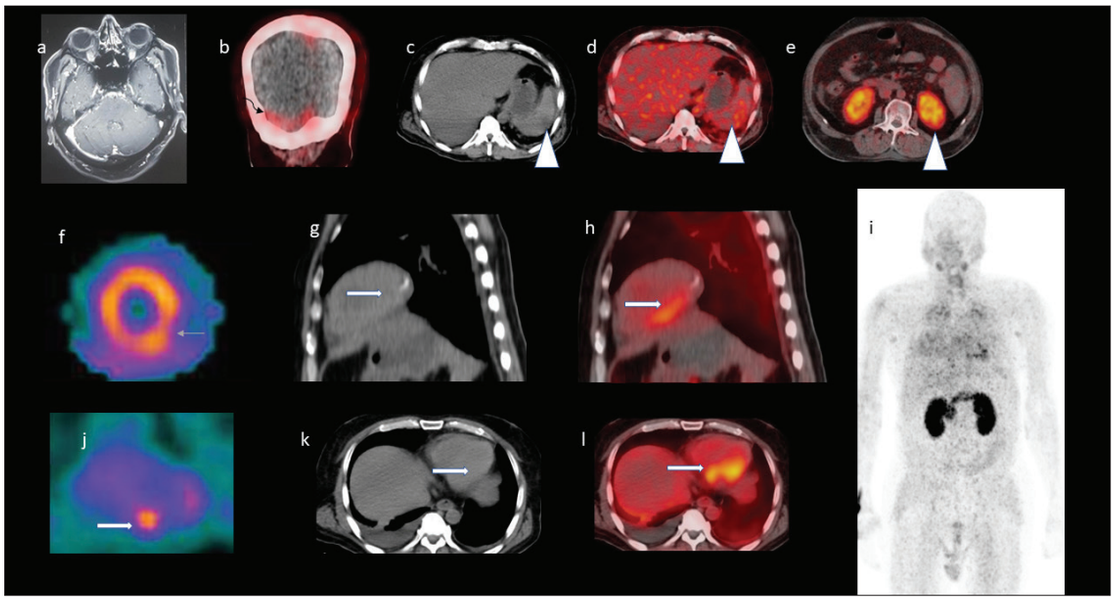



**Fig. 5:** (a-l) A 62-year-old man presented with fever for 2 months. He had altered sensorium for a week. WBC was 7600/ml. There was history of angioplasty to RCA and Circumflex coronary arteries 4 months earlier. MRI showed diffuse leptomeningitis.(a) Clinically suspected septicemia and IE. Rest MPI showed small sized mild perfusion defect in inferolateral myocardium. (thin arrow in f). FAPI scan showed tracer localisation in inferior and inferolateral myocardium. (thick arrows g,h,j,k,l) Uptake was distal to the circumflex stent and more discordant as compared to the perfusion scan. Diffuse uptake of radiotracer in basal meninges (curved arrow b), heterogeneous increased FAPI uptake in spleen corresponding to CT hypodensity (arrowheads c,d), intense uptake in kidneys was observed.(arrowhead e) Overall suggestive of IE with multiple septic thrombo-emboli involving meninges, spleen and kidneys. (i)

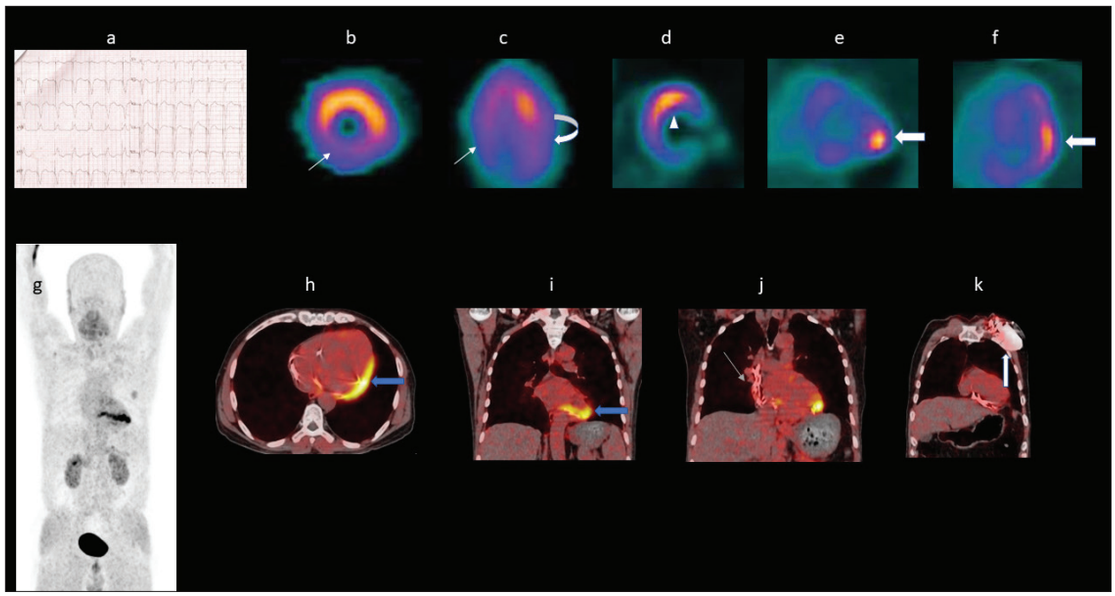



**Fig. 6:** (a-k) A 69-year-old man with dilated cardiomyopathy and ventricular failure with cardiac syndronization device developed fever. Blood culture isolated Streptococcus Hemophilus. He was on antibiotics for 4weeks. MPI revealed inferoseptal hypoperfusion (thin arrow in b, c) related to pacemaker causing bundle branch block pattern on ECG. (a) Echocardiogram showed global hypokinesia with EF30%. FDG PET scan showed metabolism in anteroseptal myocardium (arrowhead d), and was inconclusive. FAPI PET scan on next day showed linear increased FAPI uptake along the inferolateral wall of left ventricle corresponding to electrode along coronary sinus. (solid arrow e, f) Corresponding MPI images show hypoperfusion.(curved arrow c). Fusion PET scan showed FAPI localization along the electrode in coronary sinus.(solid blue arrows h, i) RV electrode (thin arrow in j) and pulse generator in left infraclavicular fossa (vertical arrow in k) did not show any abnormal tracer uptake. (g) MIP image is also shown.

**Table 1:** List of cases

| Serial number | Age/sex | Angio | MPI | FDG | FAPI | Final impression |
|---|---|---|---|---|---|---|
| 1 | 49/male | Aneurysm of LAD | Basal anterior defect | Basal septal uptake along LAD stent | NA | Mycotic aneurysm LAD |
| 2 | 55/male | Aneurysm of RCA | Apical-distal septal perfusion defect | NA | Uptake along RCA stent | Mycotic aneurysm RCA |
| 3 | 63/male | Circumflex neurysm | Anteroseptal and inferolateral defects | Uptake along circumflex stent | NA | Infected circumflex stent |
| 4 | 68/male | RCA stent | NA | NA | Along RCA stent | Infected RCA stent |
| 5 | 62/male | Circumflex stent | Inferolateral small defect | NA | Inferior wall uptake distal to stent | Infective endocarditis/ septic emboli |
| 6 | 69/male | Synchronization leads | Inferoseptal hypoperfusion | Inconclusive | Linear uptake along coronary sinus lead | Infected coronary sinus lead |

PTCA: Percutaneous transluminal coronary angioplasty, LAD: Left anterior descending, NA: Not available, MPI: Myocardial perfusion imaging, FDG: Fluorodeoxyglucose, FAPI: Fibrinogen activation protein inhibitor, RCA: Right coronary artery

CT coronary angiography (CTA) has demonstrated high sensitivity and specificity in diagnosing coronary artery mycotic aneurysms.

A retrospective analysis of 55 cases identified a lobulated contour with thick walls or mural thrombus, with or without a saccular shape, in 55% of cases. Additionally, associated pericardial fluid, thickening, or loculations were observed in 79% of cases.[5]

In our series, case 3 demonstrated evidence of pericardial reaction. Mycotic aneurysm following stent placement typically occurs secondary to transient bacteremia from skin flora, local hematoma formation, or repeated vascular access using the same sheath. In potentially life-threatening cases, surgical intervention may be required.[6]

Among the six cases in our study, three demonstrated stent-related infections/mycotic aneurysms, as identified through PET-CT imaging using a combination of MPI and FDG/FAPI PET-CT scans. Fortunately, all three responded well to aggressive antibiotic therapy and did not require surgical intervention.

A case report has previously documented a mycotic coronary aneurysm diagnosed using FDG PET.[7] In our case series, two of the three patients with stent infection were diagnosed with FDG PET-CT. The third patient with a mycotic aneurysm was diagnosed with FAPI PET-CT. One patient with CIED showed an inconclusive FDG result, which was probably related to improper preprocedural preparation. FDG shows variable uptake in the myocardium. Normal myocardium under fasting state preferentially uses free fatty acid for its metabolism. Postprandially, myocardium shifts its metabolism to glucose. This property makes FDG a preferential molecule for assessing cardiac viability in the postinfarct state to differentiate between hibernating myocardium and scar tissue. In contrast, FDG goes to the inflamed myocardium in conditions such as sarcoidosis.[8] The protocol required a prolonged fasting state and pre-administration of heparin to allow preferential utilisation of fatty acids instead of glucose. FDG uptake under such controlled conditions helps to localise inflammatory/infective areas in the myocardium. In view of patient-centric factors involved, FDG may give false positive or inconclusive results as demonstrated in case 6.[9]

A meta-analysis found FDG PET-CT to have high sensitivity and specificity in detecting infections related to CIEDs.[10] In our case series, case 6 had a cardiac resynchronisation device with evidence of infection localised to the coronary sinus lead, while the right ventricular lead and pulse generator implant site showed no abnormal uptake.

Fibroblast activation protein-alpha is a homodimeric, membrane-bound serine protease with both intracellular and extracellular soluble forms. It can be targeted using FAPI to highlight areas of inflammation and neoplasms.

Unlike FDG, FAPI does not normally localise to the myocardium. Hence, its localisation in four of the patients reported here is a promising note for the future, to make diagnosis of infected cardiac stents/devices as well as infective endocarditis.

Although FAPI uptake may occur in infarcted myocardial regions, there is currently limited literature supporting its utility in diagnosing myocardial infarction. In our case series,

cases 2 and 4 demonstrated FAPI uptake associated with myocardial infarction. However, this did not compromise diagnostic accuracy, as the suspected site of infection was located away from the infarcted regions, and PET-CT images provided anatomic correlates to confirm associated structural changes.

Clinical red flags for infective endocarditis include high fever, a new heart murmur, conduction disturbances, embolic phenomena, or cardiogenic shock due to valvular regurgitation, which should raise suspicion for infective endocarditis. Several conditions increase the risk of developing infective endocarditis, including congenital heart disease, valvular heart disease, drug-eluting stents, and CIEDs.[11] Other contributing factors include haemodialysis, prosthetic heart valves, venous catheters, immunosuppression, and CIED-related infections. Multimodality criteria are used to make the diagnosis of infective endocarditis. These include paravalvular lesions / leak on CT, abnormal FDG uptake around the prosthetic valve (beyond 3 months of implantation) or radiolabelled white blood cell SPECT CT scan. Recent embolic events and an infected aneurysm on imaging are considered minor criteria. TEE has a sensitivity and specificity of 90%–100% for native valve endocarditis, respectively.[12] Vegetations are typically seen on the low-pressure side of the valve and, when large, may embolize. Other criteria for infective endocarditis on echocardiography include paravalvular leak, abscess, new dehiscence of prosthetic valve, or new regurgitation. Complications such as valve prolapse, perforation, chordae, or papillary muscle rupture can also be diagnosed on echocardiography. Abscess tends to occur in the aortic or prosthetic valve and may be complicated by pseudoaneurysm and fistula formation.[13] FDG PET-CT has its advantages and limitations. The absence of FDG uptake does not rule out infective endocarditis. However, its superiority over conventional CT and echocardiography in prosthetic valve endocarditis is well established.[14,15] Three clinical situations to consider FDG are native valve endocarditis, prosthetic valve endocarditis, and cardiac device-related endocarditis. The sensitivity of FDG in native valve endocarditis is low (35%), and the specificity is high (99%).[16] The sensitivity and specificity are high in prosthetic valve endocarditis (86% / 84% respectively) as well as for cardiac device related infections (87 / 94% respectively).

### Advantages and limitations of fluorodeoxyglucose positron emission tomography–computed tomography and fibroblast activation protein inhibitor positron emission tomography–computed tomography

FDG PET-CT is advantageous in detecting endocardial infections and systemic sites of infection. However, FDG exhibits physiologic uptake in the kidneys, brain, and gut, which may interfere with distinguishing pathological uptake at these sites.[17]

In contrast, FAPI PET-CT does not localise to these organs, making it particularly useful in identifying systemic involvement. In our case series, FAPI uptake in the myocardium, kidneys, spleen, and meninges (as demonstrated in case 5) suggests a broader inflammatory or infectious process.[18,19]

## CONCLUSION

Myocardial stent- or device-related infections, as well as infective endocarditis, can be diagnosed or suspected using nuclear medicine imaging techniques, particularly FDG and FAPI PET-CT scans.

MPI provides additional diagnostic insights, helping to localise coexisting infarction.

While FDG PET-CT has certain limitations, these can be overcome with FAPI PET-CT, offering a promising alternative in cardiac infection imaging.

**Ethical approval:** Institutional Review Board approval is not required.
**Declaration of patient consent:** The authors certify that they have obtained all appropriate patient consent forms. In the form, the patients have given their consent for their images and other clinical information to be reported in the journal. The patients understand that their names and initials will not be published and due efforts will be made to conceal their identity, but anonymity cannot be guaranteed.
**Financial support and sponsorship:** Nil
**Conflicts of interest:** There are no conflicts of interest.
**Use of artificial intelligence (AI)-assisted technology for manuscript preparation:** The authors confirm that there was no use of artificial intelligence (AI)-assisted technology for assisting in the writing or editing of the manuscript, and no images were manipulated using AI.

## REFERENCES

1. Kung BT, Seraj SM, Zadeh MZ, Rojulpote C, Kothekar E, Ayubcha C, *et al.* An update on the role of (18)F-FDG-PET/CT in major infectious and inflammatory diseases. Am J Nucl Med Mol Imaging 2019;9:255-73.
2. Kawsara A, Núñez Gil IJ, Alqahtani F, Moreland J, Rihal CS, Alkhouli M. Management of coronary artery aneurysms. JACC Cardiovasc Interv 2018;11:1211-23.
3. Lee WC, Chung WJ, Fang HY, Wu CJ. Coronary artery aneurysms formation within everolimus-eluting stents and bioresorbable vascular scaffolds. Int J Cardiol 2016;206:58-60.
4. Omoto T, Saito K, Kashima T, Kume M, Hosaka S, Kimura S. Mycotic aneurysm of the right coronary artery. Asian Cardiovasc Thorac Ann 2006;14:331-2.
5. Kishida K, Nakaoka H, Sumitsuji S, Nakatsuji H, Ihara M, Nojima Y, *et al.* Successful surgical treatment of an infected right coronary artery aneurysm-to-right ventricle fistula after sirolimus-eluting stent implantation. Intern Med 2007;46:865-71.
6. Crook BR, Raftery EB, Oram S. Mycotic aneurysms of coronary arteries. Br Heart J 1973;35:107-9.
7. Restrepo CS, Gonzalez TV, Baxi A, Rojas CA. Infected

("Mycotic") coronary artery aneurysm: Systematic review. J Cardiovasc Comput Tomogr 2020;14:e99-104.
8. Pacella S. Mycotic coronary aneurysm detected by 18F-FDG PET/CT: A case report. J Nucl Cardiol 2021;28:3088-9.
9. Vasantrao SS, Mahaveer PA, Vasant SS, Vishnu SD. Pericardial sarcoidosis presenting as PUO diagnosed on FDG PET CT scan. Asia Ocean J Nucl Med Biol 2020;8:74-8.
10. Scholtens AM, Verberne HJ, Budde RP, Lam MG. Additional heparin preadministration improves cardiac glucose metabolism suppression over low-carbohydrate di*et al*one in $^{18}$F-FDG PET imaging. J Nucl Med 2016;57:568-73.
11. Erba PA, Lancellotti P, Vilacosta I, Gaemperli O, Rouzet F, Hacker M, *et al.* Recommendations on nuclear and multimodality imaging in IE and CIED infections. Eur J Nucl Med Mol Imaging 2018;45:1795-815.
12. Juneau D, Golfam M, Hazra S, Zuckier LS, Garas S, Redpath C, *et al.* Positron emission tomography and single-photon emission computed tomography imaging in the diagnosis of cardiac implantable electronic device infection: A systematic review and meta-analysis. Circ Cardiovasc Imaging 2017;10:e005772.
13. Li JS, Sexton DJ, Mick N, Nettles R, Fowler VG Jr., Ryan T, *et al.* Proposed modifications to the Duke criteria for the diagnosis of infective endocarditis. Clin Infect Dis 2000;30:633-8.
14. Cahill TJ, Prendergast BD. Infective endocarditis. Lancet 2016;387:882-93.
15. Doherty JU, Kort S, Mehran R, Schoenhagen P, Soman P. ACC/AATS/AHA/ASE/ASNC/HRS/SCAI/SCCT/SCMR/STS 2017 appropriate use criteria for multimodality imaging in valvular heart disease: a report of the American College of Cardiology Appropriate Use Criteria Task Force, American Association for Thoracic Surgery, American Heart Association, American Society of Echocardiography, American Society of Nuclear Cardiology, Heart Rhythm Society, Society for Cardiovascular Angiography and Interventions, Society of Cardiovascular Computed Tomography, Society for Cardiovascular Magnetic Resonance, and Society of Thoracic Surgeons. J Am CollCardiol 2017;70:1647-72.
16. Ten Hove D, Slart RH, Sinha B, Glaudemans AW, Budde RP. (18)F-FDG PET/CT in infective endocarditis: Indications and approaches for standardization. Curr Cardiol Rep 2021;23:130.
17. Sarrazin JF, Philippon F, Tessier M, Guimond J, Molin F, Champagne J, *et al.* Usefulness of fluorine-18 positron emission tomography/computed tomography for identification of cardiovascular implantable electronic device infections. J Am Coll Cardiol 2012;59:1616-25.
18. Kamani CH, Allenbach G, Jreige M, Pavon AG, Meyer M, Testart N, *et al.* Diagnostic performance of (18)F-FDG PET/CT in native valve endocarditis: Systematic review and bivariate meta-analysis. Diagnostics (Basel) 2020;10:754.
19. Jagtap RR, Savale SV, Khajindar GS, Solav SV. Localization of pyelonephritis by (68)Ga-FAPI PET CT. Asia Ocean J Nucl Med Biol 2023;11:85-8.

**How to cite this article:** Solav SV, Jagtap RR, Savale SV. Demystifying Cardiac Infections with Nuclear Medicine. Indian J Nucl Med. 2026;41:134-141. doi: 10.25259/IJNM_62_25